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From Chaos to Control: Programmable Crack Patterning with Molecular Order in Polymer Substrates

Hyun Kim, Mustafa K. Abdelrahman, Joonmyung Choi, Hongdeok Kim, Jimin Maeng, Suitu Wang, Mahjabeen Javed, Laura K. Rivera-Tarazona, Habeom Lee, Seung Hwan Ko, and Taylor H. Ware**

Cracks are typically associated with the failure of materials. However, cracks can also be used to create periodic patterns on the surfaces of materials, as observed in the skin of crocodiles and elephants. In synthetic materials, surface patterns are critical to micro- and nanoscale fabrication processes. Here, a strategy is presented that enables freely programmable patterns of cracks on the surface of a polymer and then uses these cracks to pattern other materials. Cracks form during deposition of a thin film metal on a liquid crystal polymer network (LCN) and follow the spatially patterned molecular order of the polymer. These patterned sub-micrometer scale cracks have an order parameter of 0.98 ± 0.02 and form readily over centimeter-scale areas on the flexible substrates. The patterning of the LCN enables cracks that turn corners, spiral azimuthally, or radiate from a point. Conductive inks can be filled into these oriented cracks, resulting in flexible, anisotropic, and transparent conductors. This materials-based processing approach to patterning cracks enables unprecedented control of the orientation, length, width, and depth of the cracks without costly lithography methods. This approach promises new architectures of electronics, sensors, fluidics, optics, and other devices with micro- and nanoscale features.

1. Introduction

Processes that allow patterning of materials at the micro- or nanoscale enable many devices, including active electronics, and thus have enormous societal and economic impact.[1,2] The cost, form, and performance of the device are limited by the conditions necessary for these processing techniques. For example, conventional photolithography typically requires patterned light to generate user-defined microscale patterns.[3–5] More sophisticated techniques such as electron-beam lithography or dip-pen lithography can pattern nanoscale features, but these multistep scanning processes are high-cost and labor-intensive.[6,7] Thus, fabrication techniques that can produce highly organized patterns or structures at small size scales with a simple process may offer significant advantages.

In synthetic systems, cracking is normally an unwelcomed, uncontrollable, and chaotic phenomenon caused by mechanical conditions that induce material failure. However, living systems, such as the skin of crocodiles and elephants, often show examples of self-organized crack patterns caused by intrinsic anisotropy.[8,9] Inspired by nature, recent advances in crack-assisted fabrication techniques, also referred to as crack-lithography, have demonstrated how the humble crack can be

Dr. H. Kim
Sensors and Electron Devices Directorate
CCDC Army Research Laboratory
Adelphi, MD 20783, USA
E-mail: kimhyun@utdallas.edu

Dr. H. Kim, M. K. Abdelrahman, Dr. J. Maeng, S. Wang, M. Javed, L. K. Rivera-Tarazona, Prof. T. H. Ware
Department of Bioengineering
University of Texas at Dallas
Richardson, TX 75080, USA
E-mail: taylor.ware@tamu.edu

M. K. Abdelrahman, S. Wang, Prof. T. H. Ware
Department of Materials Science and Engineering
Texas A&M University
College Station, TX 77843, USA

Prof. J. Choi, H. Kim
Department of Mechanical Design Engineering
Hanyang University
Seoul 04763, Republic of Korea

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Prof. J. Choi, H. Kim
Department of Mechanical Engineering
BK21 FOUR ERICA-ACE Center
Hanyang University
Ansan 15588, Republic of Korea

M. Javed, L. K. Rivera-Tarazona, Prof. T. H. Ware
Department of Biomedical Engineering
Texas A&M University
College Station, TX 77843, USA

Prof. H. Lee
School of Mechanical Engineering
Pusan National University
Busan 46241, Republic of Korea

Prof. S. H. Ko
Department of Mechanical Engineering
Seoul National University
Seoul 08826, Republic of Korea

Prof. S. H. Ko
Institute of Advanced Machines and Design (IAMD) / Institute of Engineering Research
Seoul National University
Seoul 08826, Republic of Korea

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used to produce high-resolution (i.e., small feature size) and high-throughput (i.e., large area) patterns.[10,11] Crack-assisted fabrication is associated with relatively facile and low-cost procedures as compared with conventional lithography strategies and has been proposed for a variety of functional applications, including flexible electronics,[12–14] micro- and nanofluidics,[15,16] optical devices,[17,18] and sensors.[19–23]

The method used to induce cracking dictates the patterns of cracks that can be obtained. Relatively simple crack patterns are readily obtained by stretching or bending a brittle film supported on a deformable substrate.[14,19–25] Cracking without externally-applied stress can result from thermally-induced residual stress in glass[26] or evaporation-driven capillary stress,[18] but control of the crack pattern is somewhat limited to chaotic wavy or grid-like patterns. To enable more complexity in crack patterns, approaches that use residual thin film tensile stress on silicon wafer substrates have been investigated.[27–32] The cracking direction on these materials is dictated by the crystallographic orientations of the silicon substrates. In these cases, lithographically-defined stress modulating notches and boundaries can be used to initiate, propagate, or terminate cracking. However, the additional use of lithography to control cracking limits the value of crack-assisted fabrication as a facile fabrication strategy. Furthermore, while silicon is the standard substrate for traditional microelectronics, the ability to use highly organized crack-assisted fabrication on polymers would be enabling for applications requiring device flexibility or stretchability.[33] An overview of various cracking systems, materials, mechanisms, processes, and applications is summarized in Table S1, Supporting Information. To overcome the limitations of previous methods, new methods to control cracking should: 1) provide a high degree of directional freedom, 2) limit additional multistep stress engineering processes (e.g., mechanical strain or notch construction), 3) enable control of the length, width, depth, and density of the cracks, and 4) be compatible with flexible substrates.

Here, we describe a straightforward method to enable programmable cracking on a polymer. This method is enabled by using glassy liquid crystal polymer networks (LCNs) with photopatterned molecular order[34–36] as substrates for deposition of stressed thin films. We note that liquid crystalline materials have demonstrated their potential as patterning substrates to create oriented buckling or wrinkles,[37–39] but the capability to control cracking has not yet been described. This distinction is important as cracks can subsequently be used as a patterning tool. The patterned molecular order of the LCNs enables several innovations in crack patterning, including: 1) freely programmable crack patterns, 2) a simple one-step process that generates high-resolution, and high-throughput crack patterns, 3) tunable crack density and size, and 4) flexibility of substrates. This materials-based processing approach presents new opportunities as an alternative multiscale patterning and fabrication strategy for various applications, including smart sensors, flexible electronics, optical devices, and micro- and nanofluidics.

2. Results and Discussion

Programmed LCN substrates for crack patterning were prepared through radical polymerization of a mixture of commercially available liquid crystal monomers (RM82 and RM257), as shown in **Figure 1a**. The monomer solution with a photoinitiator (I-369) was filled by capillary force into prepared photopatterned cells. As illustrated in Figure 1b, to prepare the photopatterned cells, a pair of glass slides were coated with an azobenzene-based dye and assembled with a gap of 37.5 µm. The assembled cell was then exposed to linearly polarized light to dictate the molecular orientation (or nematic director) of the LCN precursors, which ultimately controls the cracking direction. Briefly, azobenzene dyes can be reoriented by irradiation with linearly polarized blue light where the long axis of the molecule orients perpendicularly to the electric field vector of light[40–42] (Figure S1, Supporting Information). The oriented dye layer then controls the orientation of the nematic monomer solution. We note that similar materials and photopatterning procedures are already widely employed in the display industry and, as such, these processes could be readily be scaled to large dimensions.[43,44] After filling the mixture into the cell, the monomers were subsequently crosslinked in the aligned nematic state by photopolymerization, resulting in glassy LCNs with a broad glass transition temperature (T_g) around 62 °C (Figure S2a, Supporting Information). After crosslinking, the prepared LCN substrate remains on one of the glass slides for the crack patterning process. Chromium (Cr) was then sputtered (100 nm thick) on the LCN film. On removal of the film from the sputtering chamber, patterned cracks are observed along the programmed molecular orientation within the substrates, as illustrated in Figure 1c. As the sputtered Cr possesses internal tensile stresses,[45–47] cracking within the Cr layer is not unexpected. However, it has not been previously reported that cracks form along the direction of molecular orientation when deposited on LCNs. As shown in Figure S2b, Supporting Information, the measured storage modulus of the LCN substrate is 2.55 ± 0.10 GPa and 1.51 ± 0.81 GPa parallel and perpendicular to the molecular orientation, respectively. As a result of this mechanical anisotropy in LCN substrates, we hypothesized that the strain resulting from the deposition of the stressed Cr is relieved by stretching the underlying substrate to a greater degree perpendicular to the nematic director. This subsequently results in cracks along the nematic director (Figure S3, Supporting Information), with a high degree of orientation order parameter 0.98 ± 0.02. The order parameter, S, is calculated by $S = \cos(2\theta)$ where θ denotes the angle of cracks with respect to the nematic director, and $S = 1$ indicates perfect alignment.[48]

Patterned cracking can be realized by patterning multiple domains of nematic director within the same sample, as shown in Figure 1d. In this case, we program a sample with domains that read "LC". Outside of the letters, the nematic director is along the horizontal axis of the photo, while within the letters, the nematic director is at 45° to the horizontal axis. Within domains, the cracks are highly aligned. Individual cracks propagate between these domains turning with the change in molecular orientation. The molecular orientation of the underlying LCN substrate results in birefringence that can be observed through the cracks, with bright cracks inside of "LC" and dark cracks in the background, when observed between crossed polarizers. After the crack formation process, the deposited Cr can be easily removed with a conventional chemical etchant. As evident in Figure 1e, cracks penetrate the interface and

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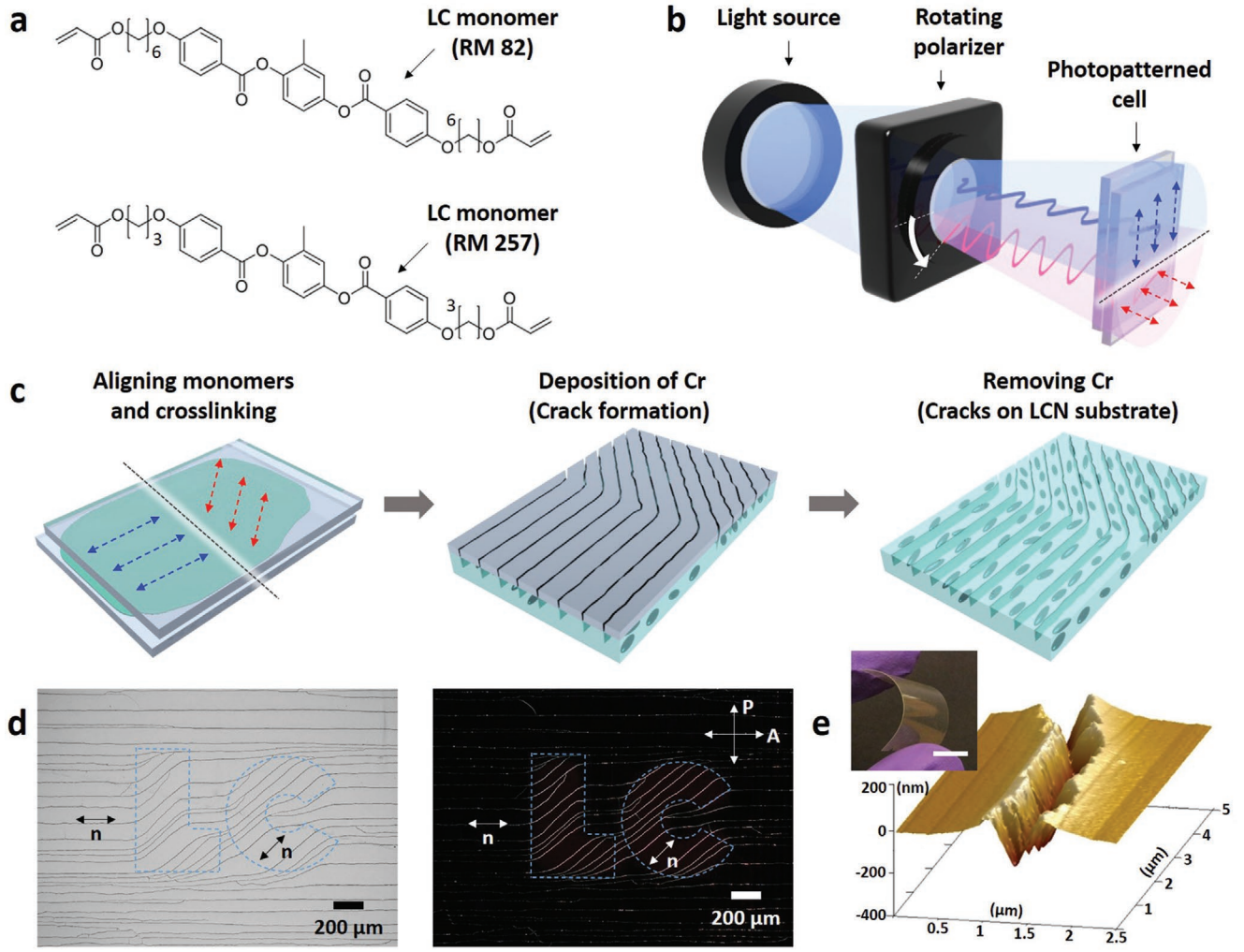


Figure 1. Overview of crack patterning with LCNs. a) Molecular structures of the liquid crystal monomers used to prepare the substrates. b) Illustration of photopatterning the cells used to make the LCN. Blue and red waves indicate angle of linear polarized blue light with rotated polarization. The blue and red arrows indicate photopatterning direction of coated azobenzene-based dye by sequential exposure, after changing the polarization angle of the incoming light. c) Illustration of crack patterning procedure. The blue and red arrows indicate the molecular orientation of LCN substrates. The ellipses indicate mesogens. d) Representative reflection optical microscopy (left) and POM (right) images of a sample with patterned cracks at 45° to the horizontal axis within "LC" and with patterned cracks at 0° to the horizontal axis in the background after deposition of Cr on the programmed LCN substrates. The defined molecular orientation is denoted by "n" (nematic director) in the images. e) Representative AFM after removing Cr showing a crack at the surface of the LCN. The inset optical image shows a free-standing and flexible LCN film with patterned cracks at the surface. (inset scale bar: 5 mm)

propagate within the surface of the LCN substrates. This propagation from the deposited thin film to the LCN substrate is related to the relative compliance of the polymer as compared to the Cr.[49] Sub-micrometer-scale cracks (181 ± 38 nm depth and 412 ± 54 nm width) are found at the surface of LCN substrates. These LCNs with patterned cracks can then be released from the carrier glass slide and used as free-standing, flexible materials (Figure 1e inset).

To understand the cracking of Cr thin films deposited on LCNs, all-atom molecular dynamics (MD) simulations were used (**Figure 2a**, Figure S4, and Table S2, Supporting Information). The modeled LCN substrate shows an anisotropic mechanical response (Figure S5, Supporting Information), indicating that the simulation environment replicated the key component of the experimental conditions. As shown in Figure 2b,c, the Cr atoms were then deposited from the top of the modeled LCN substrate for 3.5 ns total simulation time. We define a 1 nm-thick region at the top of the modeled LCN substrate to quantify the residual stress in the interfacial LCN layer. In the 1 nm-thick LCN interfacial region, anisotropic residual compressive stresses (denoted as negative) were observed: –44.6 MPa perpendicular to the director (x-axis) and –14.7 MPa parallel to the director (y-axis). As expected, compressive stress is observed in the interfacial layer of the substrates, which is induced by the tensile stress of the thin film (Figure S6, Supporting Information). Critically, the magnitude of the stress components applied to the LCN interfaces is anisotropic. In other words, stronger compressive stress is exerted in the perpendicular direction to the nematic orientation, which in turn should lead to larger deformation perpendicular to the nematic

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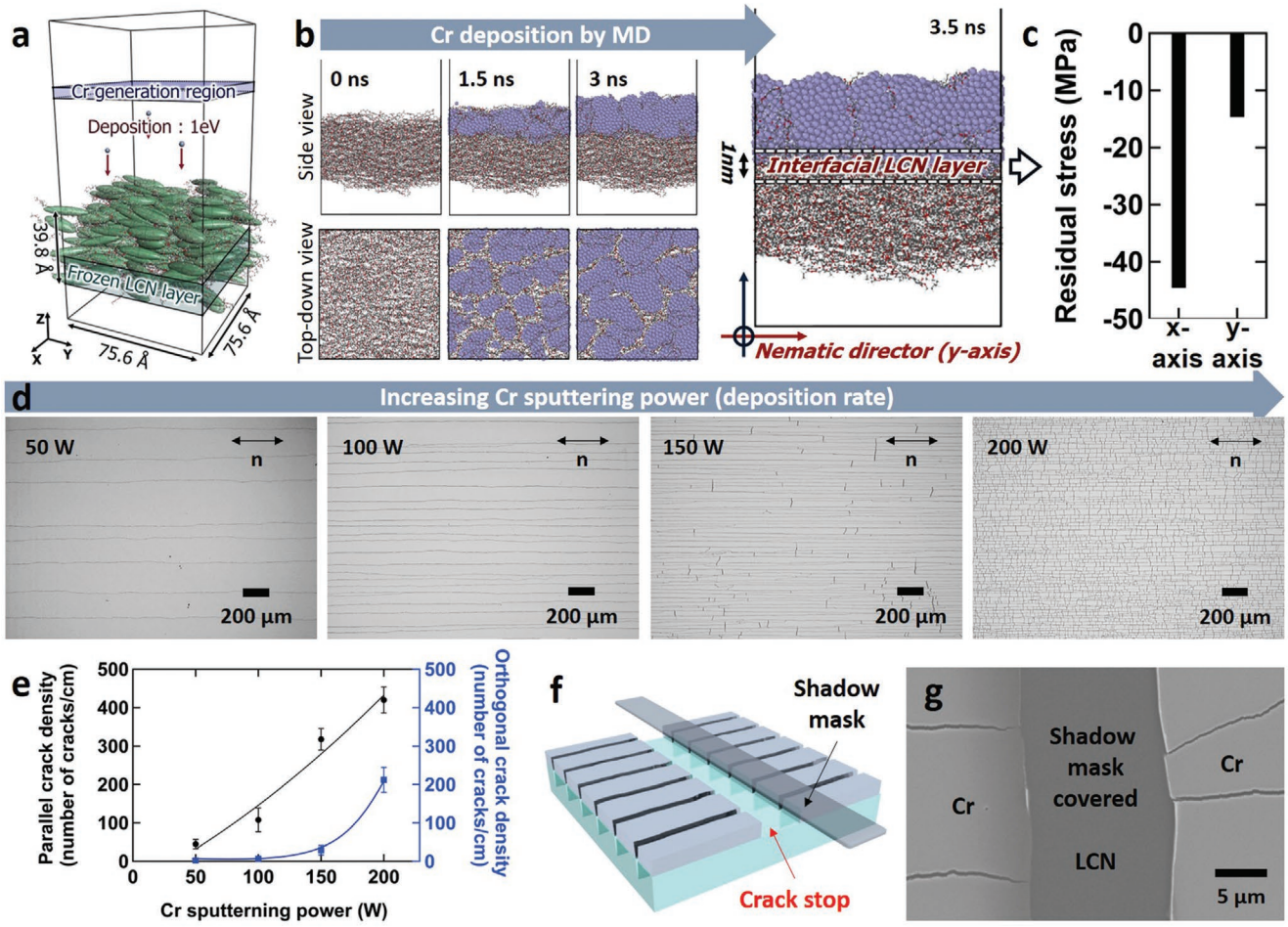


Figure 2. Crack patterning mechanism, density control, and termination. a) MD simulation model for Cr deposition on the LCN substrate. The violet circles and green ellipses represent Cr atoms and mesogens, respectively. b) Morphology variation of the LCN substrate and deposited Cr with respect to elapsed time. c) Anisotropic residual stress in a 1 nm-thick interfacial LCN layer after deposition completion. d) Reflection optical microscopy images showing control of crack density by increasing Cr sputtering power from 50 to 200 W, with a fixed Cr thickness of 100 nm. e) Crack density parallel and orthogonal to the nematic director as a function of Cr sputtering power. Trend lines are added to guide the eye. f) Illustration and g) SEM image showing crack termination enabled by the use of a shadow mask during Cr sputtering.

director. This MD simulation results indicate anisotropic interaction between the deposited Cr atoms and the oriented LCN substrate, hence, support the experimental observation of the cracking direction being parallel to the molecular orientation.

As discussed so far, the tensile stress within the Cr thin film is the driving force for crack formation. Thus, we varied the sputtering parameters, which are known to affect the thin film stress, to provide further control of the cracking. As expected, higher sputtering power resulted in faster deposition and higher tensile stress (Figure S7, Supporting Information). A series of optical microscopy images are presented in Figure 2d that show crack morphology with increasing sputtering power from 50 to 200 W, with a fixed Cr thickness of 100 nm. Increased sputtering power results in denser cracks, as summarized in Figure 2e. The parallel crack density (i.e., number of cracks along the nematic orientation) roughly linearly increases from 45 ± 12 to 420 ± 34 cracks cm^{-1} by increasing the Cr sputtering power from 50 to 200 W. For samples prepared with low sputtering powers (50 or 100 W), orthogonal cracks (i.e., cracks perpendicular to the nematic orientation) were rarely found. The density of these orthogonal cracks was 2.5 ± 0.7 and 7.2 ± 4.9 cracks cm^{-1} for 50 and 100 W, respectively. At 150 and 200 W sputtering power, these orthogonal side cracks are widespread with 29.1 ± 12.4 and 212.1 ± 32.4 cracks cm^{-1}, respectively. This experimental observation, increased crack density by higher tensile stress, is expected,[50] as the residual tensile stress affects global strain in the sample. The transition from very few to common orthogonal cracks in films prepared with 150 and 200 W sputtering powers can be explained by the increase in the magnitude of the tensile stress under these conditions. While the tensile stress in all of the Cr films is biaxial, the overall biaxial stress is relieved with cracks in a single direction when the Cr is deposited at lower powers. For higher sputtering powers, the magnitude of the biaxial tensile stress increases such that the failure strain of the Cr is reached in both directions. We also note that at higher stresses the cracks along the orientation of the polymer become closer, which can lead to crack-to-crack interactions[51] that also cause orthogonal cracking. For the rest of the study, we fixed

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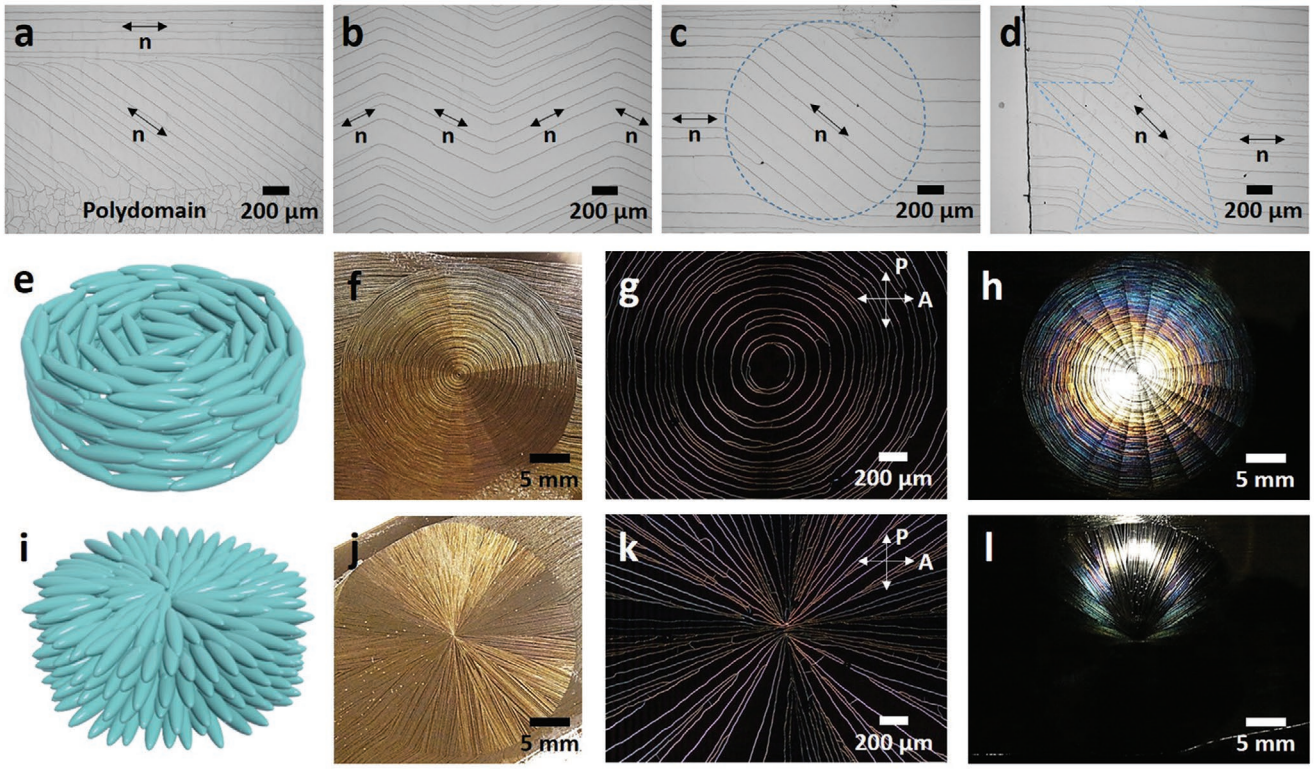


Figure 3. Complex patterning of cracks. a) Reflection optical microscopy images for LCN with cracks patterned by polydomain alignment (bottom), at 45° to the horizontal direction (middle), and 0° to the horizontal direction (top). b) Cracks running through multiple domains with a refraction angle of 120°. c,d) Patterned cracks at 45° to the horizontal direction within a circle (c) and star (d), with cracks at 0° to the horizontal direction in the background. e,i) Illustrations of azimuthal (e) and radial (i) alignment. f,j) Optical images of the patterned cracks in the azimuthal (f) and radial (j) directions. g,k) POM images showing birefringence of the substrates through the cracks at the center of azimuthal (g) and radial (k) patterns. h,l) Optical images of diffraction induced by azimuthal (h) and radial (l) cracks. The backlight (white light) is located at the center for azimuthal cracks and at the top for radial cracks, respectively.

the Cr sputtering power as 100 W to minimize unwanted orthogonal side cracking and crack-to-crack interaction. It is also noted that variation of deposited Cr thickness can provide further control of initial crack dimensions, as shown in Figures S8 and S9, Supporting Information. This effect is similar to the cracking depth control of previously reported thin-film metals on polymer substrates.[52] For example, 50 nm Cr films induce cracks that are less than 100 nm deep in the polymer film. At larger thicknesses of the Cr film, the crack depth, and width increase. When depositing 25 nm Cr, cracks are limited to only the Cr layer and do not penetrate into LCN substrate. The lack of crack propagation from the Cr film to the LCN at small thicknesses is likely due to the relatively higher compliance of the thinner Cr film.[49] With further tuning of the substrate mechanical properties and processing parameters, the ability to control crack size, density, and patterns in our system indicates promising potential as an alternative strategy for micro- and nanoscale fabrication.

As propagating cracks typically continue to a free boundary, strategies to arrest cracks (i.e., crack stop or crack termination) are critical to the utility of this patterning approach. In prior work on crack-assisted fabrication, crack termination is normally achieved by patterning structures on the substrate before cracking, such as stair-profiled structures[32,53] or notch-to-notch termination sites.[30,31] These methods require additional multistep lithography processes for structural modification of the substrates. In this work, cracking results from stresses arising from Cr deposition, and as such, cracks do not form in areas of the sample without Cr deposition. By using a shadow mask during deposition, cracks can be terminated at regions behind the mask, as shown in Figure 2f,g. Notably, additional lithography processes are not required. To understand the crack termination mechanism in our system, an MD simulation model is described in Figure S10, Supporting Information with the associated text. Briefly, the stress field between the LCN substrate and deposited Cr film can be locally controlled (i.e., no Cr coated leads to no stress generated under the shadow mask), and thus, cracks can be immediately stopped.

So far, we have discussed the fundamentals of LCN-based crack patterning. It should be emphasized that the salient feature of our system is the freely programmable nature of liquid crystalline order in the substrates. Thus, crack patterns can also be freely programmable. In **Figure 3a**, the cracking patterns are shown that result from Cr deposition on a sample with three distinct regions, two oriented domains with the nematic director between the regions varied by 45° and a third polydomain region. The oriented domains are characterized by oriented cracks, while the polydomain region is characterized by random cracks. In samples with multiple oriented domains and distinct orientations, individual cracks turn along the patterned

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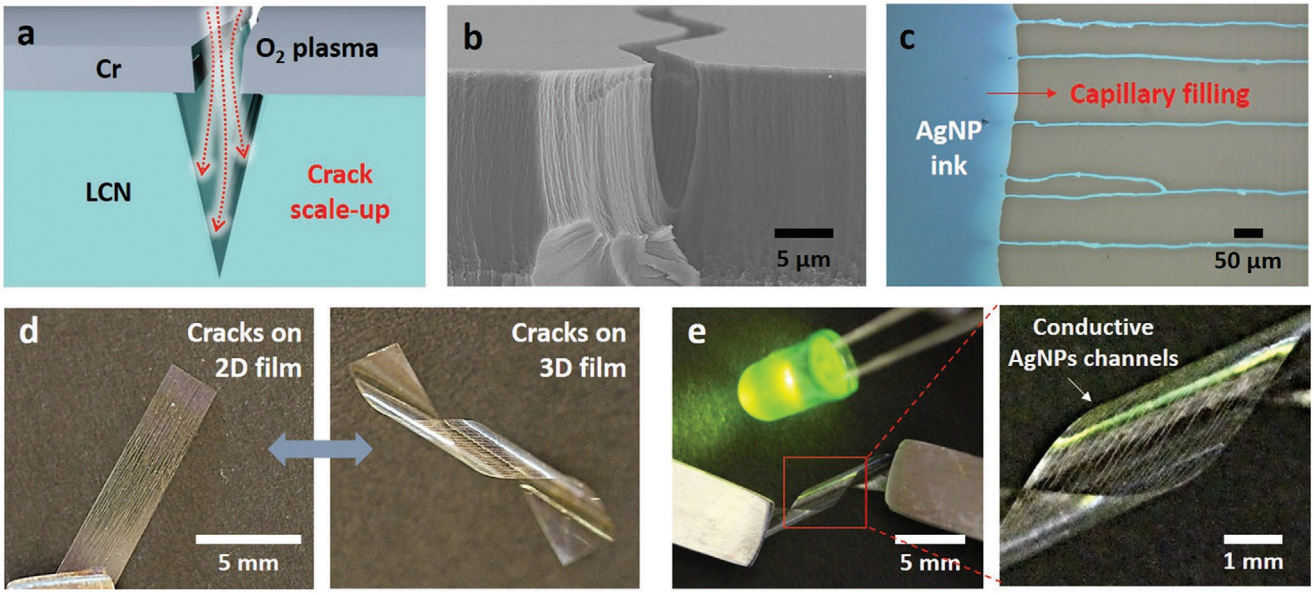


Figure 4. Post-processing of cracks. a) Illustration of post-processing for crack scale-up. The red arrows indicate O_2 plasma passed through the cracks. b) Cross-sectional SEM image of a crack after 1800 s O_2 plasma exposure and subsequent Cr removal. c) Reflection optical microscopy image showing AgNP ink filled into the scaled-up cracks by capillary force. d) Optical images of shape transformation of LCN/AgNP film. e) Optical images showing LED lighting when in series with the 3D helical LCN/AgNP anisotropic conductor.

nematic orientation within the underlying LCN substrates as the crack traverses these domains (Figure 3b). More complex spatial crack patterns are readily formed, such as circle or star-shaped domains, as shown in Figure 3c,d and Figure S11, Supporting Information. Moreover, azimuthal and radial crack patterns can be generated using corresponding patterns of molecular orientation. We note that these continuous patterns would be particularly difficult to achieve with other crack patterning approaches, such as mechanical straining or cracking along particular crystallographic directions of a wafer. Both azimuthal and radial patterns are shown in 3 cm diameter circles (Figure 3e–g,i–k and Figure S12, Supporting Information). As a result of the high-resolution (cracks with sub-micrometer scale dimensions) and high-throughput (cracks over macroscopic area) azimuthal and radial cracking, complex diffraction gratings that manipulate transmitted light in different ways are readily fabricated (Figure 3h,l).

After cracking, the sample can be post-processed to further control the dimensions of the cracks. Cracks can be scaled-up in size by oxygen (O_2) plasma reactive ion etching (RIE), as illustrated in **Figure 4a**. After crack formation, the sputtered Cr film protects the underlying LCN substrate during RIE (i.e., Cr film is a hardmask). As a result, only the cracked area of the LCN substrate can be etched. Initial sub-micrometer scale cracks can be expanded in depth and width from 181 ± 38 nm to 13.1 ± 0.4 µm and from 412 ± 54 nm to 3.2 ± 0.3 µm, respectively, after 30 min of O_2 plasma RIE (Figure 4b and Figure S13, Supporting Information). This crack scale-up process can be used to enable flow within the cracks, which can allow for the crack pattern to be applied to other materials. For example, a viscous silver nanoparticle (AgNP) ink does not flow in the initially formed sub-micrometer cracks. After the crack scale up, the AgNP ink spontaneously filled along the cracks, as shown in Figure 4c.

As the LCN substrates are polymer networks, the cracks largely maintain their form after heating and macroscopic deformation of the material. To demonstrate this point, the films are heated through the T_g (62 °C), deformed, and then fixed by cooling down to room temperature. As is well known in shape memory polymers, a deformed structure can recover its original shape by heating again up to the T_g. With this approach, substrates with patterned cracks can also be used as 3D structures, which would be difficult with most other crack patterning systems (e.g., cracks on rigid silicon wafers). To show this combination of properties, we use a cracked LCN to template a conductive material. Specifically, a uniaxially aligned with scaled-up cracks LCN was filled with AgNP ink. The resulting device is a highly-anisotropic conductor. This sheet has a surface resistivity of 58 ± 12 Ω sq^{-1} along the cracking direction, while no conductivity is measured in the orthogonal direction. This composite film was then transformed into a strain tolerant (i.e., stretchable and bendable) 3D helical shape[54] using the shape memory effect (Figure 4d). A light-emitting diode (LED) can be turned on when the LCN/AgNP is connected in series with a power supply and the LED if the connection is made along the nematic director/cracking direction, as shown in Figure 4e. If the LCN/AgNP is connected normally to the nematic director/cracking direction, the LED cannot be operated, as expected (Figure S14 and Figure S15, Supporting Information). As the patterned multiscale cracks can be filled with other functional materials, this anisotropic, transparent conductor (Figure S16, Supporting Information) is just one potential application of the processing approach described in this work.

3. Conclusion

We report a strategy to pattern cracks in a material with unprecedented control of crack direction. The simple strategy described consists of a top-down process, Cr sputtering, and

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bottom-up material design, molecular programming of LCN substrates. Through the programmed spatial variation in molecular anisotropy within the substrates, sub-micrometer-scale cracks can be freely patterned over a macroscopic area without any additional stress engineering processes. These cracks can then be scaled to larger dimensions and filled with functional materials. In addition, the programmable and processable cracks can be further extended with various types of tunable liquid crystalline polymer substrates, including liquid crystal elastomers[35,55] (Figure S17, Supporting Information). The described approach promises a potential alternative to conventional lithography techniques where processing costs and requirements limit the devices that can be made. We envision that these advances in crack-lithography will be leveraged for applications in electronics, sensors, fluidics, optics, and beyond.

4. Experimental Section

Materials: The liquid crystal monomer, 1,4-bis-[4-(6-acryloyloxyhexyloxy) benzoyloxy]-2-methylbenzene (RM82) and 1,4-bis-[4-(3-acryloyloxypropyloxy) benzoyloxy]-2-methylbenzene (RM257) were purchased from Wilshire Technologies. The photoinitiator, Irgacure I-369, was donated by BASF. A photoalignable dye, brilliant yellow, was purchased from Sigma Aldrich. All materials were used as received unless otherwise noted.

LCN Substrates Preparation: Glass slides (Electron Microscopy Sciences) were cleaned by sonication for 10 min each in detergent (Alconox), acetone (Fisher Scientific), and isopropanol (Fisher Scientific), then subsequently dried with nitrogen. The cleaned glass slides were exposed to O_2 plasma (Trion Technology, Sirius T2) for 1 min at 100 mTorr pressure and 50 W power. 1 wt% of a photoalignable dye, brilliant yellow (Sigma Aldrich), was dissolved in dimethylformamide (Fisher Scientific), then filtered by a 0.45 µm filter (Whatman). The prepared brilliant yellow solution was then spin-coated on to the prepared glass slides at 750 rpm for 10 s and at 1500 rpm for 30 s. A pair of brilliant yellow coated glass slides were assembled with 37.5 µm spacers (Precision Brand), then exposed to linearly polarized broadband light at an intensity of 10 mW cm^{-2} (modified Vivitek, D912HD) for 2 min with desired illumination pattern.[56] An equimolar amount of RM82 and RM257 with 1.5 wt% of I-369 were mixed by heating and vortexing. The equimolar mixture of two slightly different monomers was used here to prevent quick crystallization before crosslinking. The prepared monomer mixture was filled into a prepared photopatterned cell by capillary action at 70 °C. After filling, the aligned monomers were cooled at room temperature for 1 min, then exposed to ultraviolet light at an intensity of 250 mW cm^{-2} (Lumen Dynamics, OmniCure LX400+) for 5 min for crosslinking. After crosslinking, the top-side glass slide was removed for Cr sputtering.

LCN Substrates Characterization: Thermal properties were measured by a differential scanning calorimetry (DSC) (TA Instruments, DSC 2500). Samples of at least 1 mg were loaded into aluminum pans and heated to 150 °C, cooled to −50 °C, and then heated again to 150 °C with ramps of ±10 °C min^{-1}. Mechanical properties were measured using a dynamic mechanical analyzer (TA Instruments, RSA-G2). Test specimens were prepared with dimensions of approximately 10 mm × 2 mm × 37.5 µm (length × width × thickness) with the long axis parallel or perpendicular to the nematic direction. Storage modulus was measured by sweeping oscillation amplitude 0.01–1.0% of logarithmic sweep strain at 1 Hz frequency at room temperature. All tests were run in triplicate.

Crack Patterning: 100 nm of Cr was deposited on a programmed LCN substrate via sputtering system (AJA International, ATC Orion) with a radio frequency magnetron gun at the desired sputtering power. To observe crack dimensions as a function of Cr thickness, 25, 50, 150, and 200 nm of Cr was deposited at 100 W of sputtering power. The vacuum chamber pressure was maintained at 4 mTorr with 12 sccm Argon (Ar) flow rate at room temperature. A 25 µm-thick polyester film (Precision Brand) was used as a shadow mask for experiments with crack termination. To remove Cr, samples were immersed in a Cr-etchant for 5 min (KMG Electronic Chemicals). A profilometer (Veeco, Dektak 8) was used to measure the Cr film thickness. A thin film stress measurement system (Toho Technology, FLX-2320-S) was used to measure the Cr film stress when deposited on a silicon wafer (University wafer, 100 mm).

Crack Characterization: Optical microscopy images were taken with a polarized optical microscope (POM) (Olympus, BX51). Macroscopic images were taken using a digital single-lens reflex camera (Cannon). Crack dimensions were measured using an atomic force microscope (AFM) (Veeco, Dimension 500 SPM) and scanning electron microscope (SEM) (Zeiss, Supra 40). For cross-sectional observation, samples were cryofractured in liquid nitrogen. The LED of a cell phone (Samsung, Galaxy S8) was used as a backlight for diffraction grating observation. The absorption spectra were measured by a UV–Vis spectrophotometer (Agilent, Cary 5000). Three samples were tested.

Crack Post-Processing: To scale-up crack dimensions, cracked LCNs with Cr layer were exposed to O_2 plasma RIE (Trion Technology, Sirius T2) for the desired time at 150 mTorr pressure and 150 W power. After the O_2 plasma etching, the sample was immersed into a Cr etchant to remove the Cr layer. For conductivity patterning, a drop of AgNP ink (Harima Chemicals, NPS-J) was placed at the edge of a cracked and LCN film with 1800 s RIE etching for capillary filling. After filling, the AgNP was sintered at 175 °C for 30 min. For shape transformation, the LCN/AgNP film was heated up to 125 °C and deformed, then cooled down to room temperature to fix the deformed shape. Surface resistivity was measured using a multimeter (Fluke 114) with liquid metal, eutectic Gallium-Indium (EGaIn), contacts. For LED lighting, the LCN/AgNP film was connected to a DC power supply (Agilent, E3612A) with EGaIn electrical contacts.

LCN Substrate Modeling for MD Simulation: The LCN substrate was modeled by all-atom MD simulations. Initially, the same number of the RM257 and RM82 molecules were randomly located in the unit cell under a low-density condition of 0.2 g cc^{-1}. To describe the bulk material system, periodic boundary conditions were applied to all directions of the unit cell. The aligned nematic phase of the LCN substrate was obtained as the mesogens of the liquid crystal monomers were aligned along the y-axis. Then, the unit cell under low density was compressed until the density reached the target value of 1.0 g cc^{-1}, while maintaining the alignment of mesogens. At each compression stage, the potential energy of the system was optimized by a conjugate gradient method with a cutoff of 10 kcal mol^{-1} $Å^{-1}$. The polymer consistent force field (PCFF) was used to describe the interatomic potential energy of the LCN model.[57] For the optimized microstructure, the NVT ensemble (300 K) for 1 ns and the NPT ensemble (300 K, 1 atm) for 1 ns were sequentially conducted using the Verlet velocity algorithm with a time step of 1 fs. Subsequently, crosslinking of the LCN was performed. The distances between the reactive sites of the monomers were measured, and covalent bonds were created when these distances were shorter than the cutoff distance. The cutoff distance started from 4 Å, then steadily increased at intervals of 0.5 Å until 75% of the monomers were polymerized. Finally, the polymerized LCN substrate was further equilibrated using the NVT ensemble (300 K) for 1 ns and the NPT ensemble (300 K, 1 atm) for 2 ns. The all-atom modeling and the equilibration procedure of the LCN model were carried out through a commercial MD simulation package, Materials Studio (Dassault Systemes, 2020).

Deriving Mechanical Properties of the Modeled LCN Substrate: Anisotropic mechanical properties of the modeled LCN substrate were obtained through the uniaxial tensile test by open-source code, LAMMPS (Sandia Lab.). Before elongation, the LCN substrate was equilibrated under the NVT ensemble with a low temperature of 0.1 K for 0.5 ns and then deformed until the strain of the system reached 0.15% in the engineering strain rate of 10^6 s^{-1}. The effective stress tensor (σ) was determined from the virial theorem and can be calculated by Equation (1) as follows:[58]

$$\sigma = \frac{1}{V}\left(-\sum_{i}^{N} m_i (v_i v_i^T) + \frac{1}{2}\sum_{i}^{N}\sum_{j\neq i}^{N} r_{ij} F_{ij} \right) \quad (1)$$

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where V is the volume of the LCN substrate, N is the number of atoms, m_i and v_i are the mass and velocity of the ith atom, respectively. In addition, r_{ij} indicates the distance vector and F_{ij} represents the interatomic force between the ith and jth atoms. Since the tensile simulation was performed at an extremely low temperature, σ can be converted directly to the Cauchy stress.[59] Therefore, the stiffness tensor (C) of the modeled LCN can be derived using the following Equation (2):

$$\sigma = C\varepsilon \tag{2}$$

where ε is the strain applied to the LCN substrate.

Cr Deposition Simulation and Residual Stress Calculation: Cr deposition simulation was performed by using an open-source code, LAMMPS (Sandia Lab.). During the deposition simulation, the Cr atoms were equilibrated under the NVE ensemble with the embedded atom method potential.[60] Simultaneously, the LCN substrate was equilibrated under the NVT ensemble (300 K) with the PCFF. The atomic velocities of the LCN constituents were rescaled to 300 K every 0.5 ps to prevent undesired kinetic energy generation in the LCN substrate. Then, the Cr generation region of 0.5 nm thickness was defined at 5 nm away from the LCN substrate. In this region, Cr atoms were generated at random locations within every 0.5 ps and were thrown in the $-z$ direction with constant energy of 1 eV. The Cr removal region was defined on the top of the Cr generation region, and the Cr atoms located in this region were periodically removed. This was for the Cr atoms reflected in the opposite direction by strong repulsive interaction with the LCN or as-deposited Cr layer. The area from the bottom of the LCN to 1.5 nm was fixed to suppress the moving of the LCN substrate. The deposition of Cr was performed until the total simulation time reached 3.5 ns. The interaction between the LCN substrate and Cr atoms was calculated using the Lennard-Jones potential (9-6) with the parameters implemented in the PCFF. To quantify the residual stress, the LCN substrate was equilibrated for 0.5 ns under NVT ensemble at a low temperature of 0.1 K while the Cr atoms maintained their NVE ensemble. The LCN constituents in the 1 nm-thick interfacial region were defined, and the residual stress tensor was calculated using the virial theorem shown in Equation (1).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

crack-assisted fabrication, crack-lithography, liquid crystal polymer networks, patterning, unconventional fabrication

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